In [1]:
```
import pandas as pd
import random

names = ["Aryan", "Rahul", "Sneha", "Anjali", "Vikram", "Priya", "Rohit", "Neha"
tasks = [f"Task {i}" for i in range(1, 21)]
status_options = ["Done", "Pending", "In Progress", "Delayed"]

data = {
    "Name": [random.choice(names) for _ in range(100)],
    "Task": [random.choice(tasks) for _ in range(100)],
    "Status": [random.choice(status_options) for _ in range(100)],
    "Hours_Spent": [round(random.uniform(1, 8), 1) for _ in range(100)],
    "Priority": [random.choice(["High", "Medium", "Low"]) for _ in range(100)]
}

df = pd.DataFrame(data)
df.head()
```

Out[1]:

	Name	Task	Status	Hours_Spent	Priority
0	Vikram	Task 20	Pending	7.7	Low
1	Rahul	Task 18	In Progress	5.0	Medium
2	Sneha	Task 16	Pending	7.9	High
3	Priya	Task 11	In Progress	7.4	High
4	Priya	Task 17	In Progress	3.1	Medium

In [2]:
```
df.to_csv("employee_task_dataset.csv", index=False)
```

In [3]:
```
import os
os.listdir()
```

Out[3]:
```
['.anaconda',
 '.android',
 '.conda',
 '.condarc',
 '.continuum',
 '.dotnet',
 '.emulator_console_auth_token',
 '.gradle',
 '.idlerc',
 '.ipynb_checkpoints',
 '.ipython',
 '.jupyter',
 '.ms-ad',
 '.python_history',
 '.vscode',
 '3D Objects',
 'AppData',
 'Application Data',
 'Contacts',
 'Cookies',
 'Documents',
 'Downloads',
 'eeeee.pdf',
 'employee_data.csv',
 'employee_task_dataset.csv',
 'Favorites',
 'IntelGraphicsProfiles',
 'Linear Regression GFG.pdf',
 'Links',
 'Local Settings',
 'My Documents',
 'My project',
 'My project (1)',
 'My project (2)',
 'NetHood',
 'Nptel Fees for Introduction to Industry 4.0 and Industrial IOT.pdf',
 'NTUSER.DAT',
 'ntuser.dat.LOG1',
 'ntuser.dat.LOG2',
 'NTUSER.DAT{e8132d60-197e-11f1-a439-c379e4aeb208}.TM.blf',
 'NTUSER.DAT{e8132d60-197e-11f1-a439-c379e4aeb208}.TMContainer00000000000000000
001.regtrans-ms',
 'NTUSER.DAT{e8132d60-197e-11f1-a439-c379e4aeb208}.TMContainer00000000000000000
002.regtrans-ms',
 'ntuser.ini',
 'OneDrive',
 'PrintHood',
 'Recent',
 'Saved Games',
 'ScStore',
 'Searches',
 'SendTo',
 'source',
 'Start Menu',
 'Templates',
 'Tracing',
 'Untitled.ipynb',
 'Untitled1.ipynb',
 'Untitled2.ipynb',
 'Untitled3.ipynb',
```

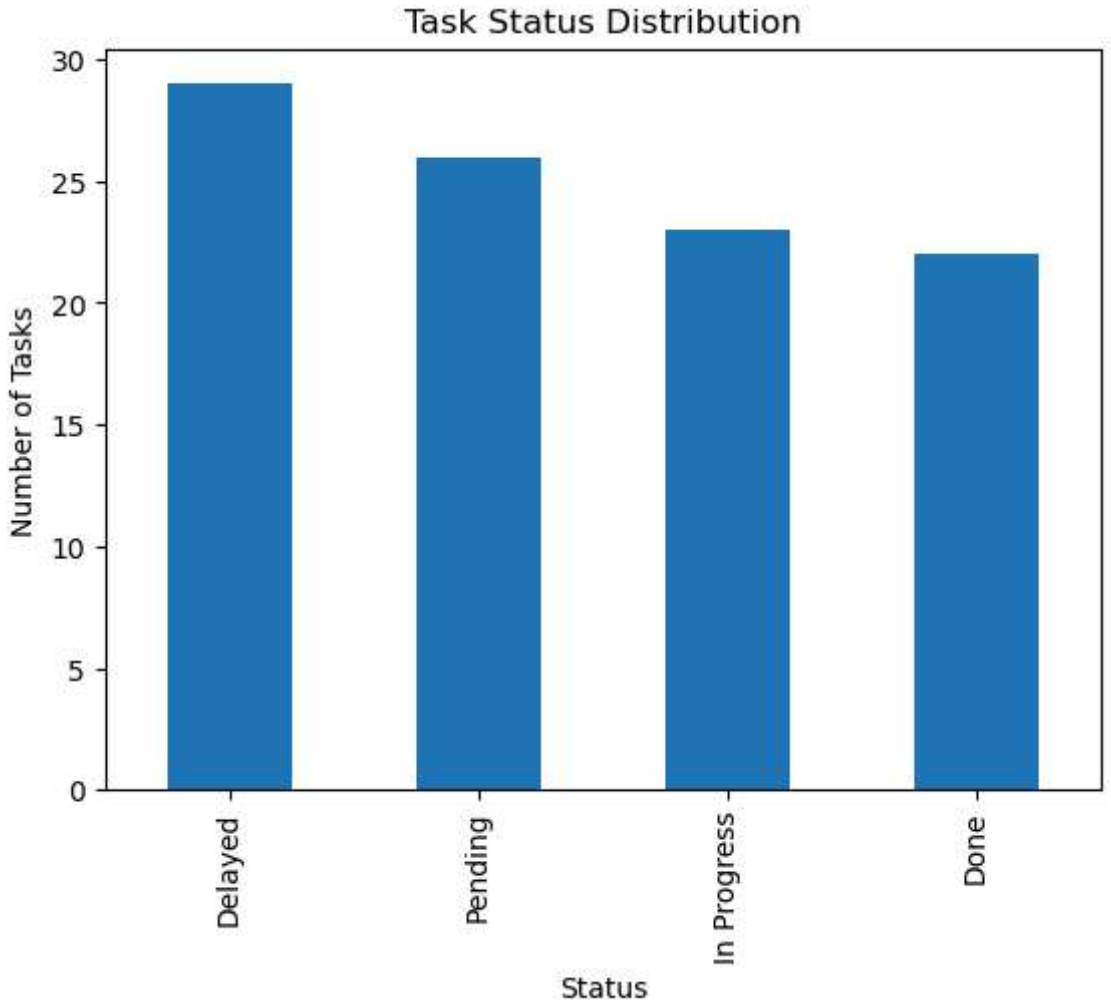
```
 'Untitled4.ipynb',
 'Untitled5.ipynb',
 'Untitled6.ipynb',
 'Untitled7.ipynb']
```

In [4]:
```
df["Status"].value_counts()
```

Out[4]:
```
Status
Delayed        29
Pending        26
In Progress    23
Done           22
Name: count, dtype: int64
```

In [5]:
```
import matplotlib.pyplot as plt

df["Status"].value_counts().plot(kind="bar")
plt.title("Task Status Distribution")
plt.xlabel("Status")
plt.ylabel("Number of Tasks")
plt.show()
```

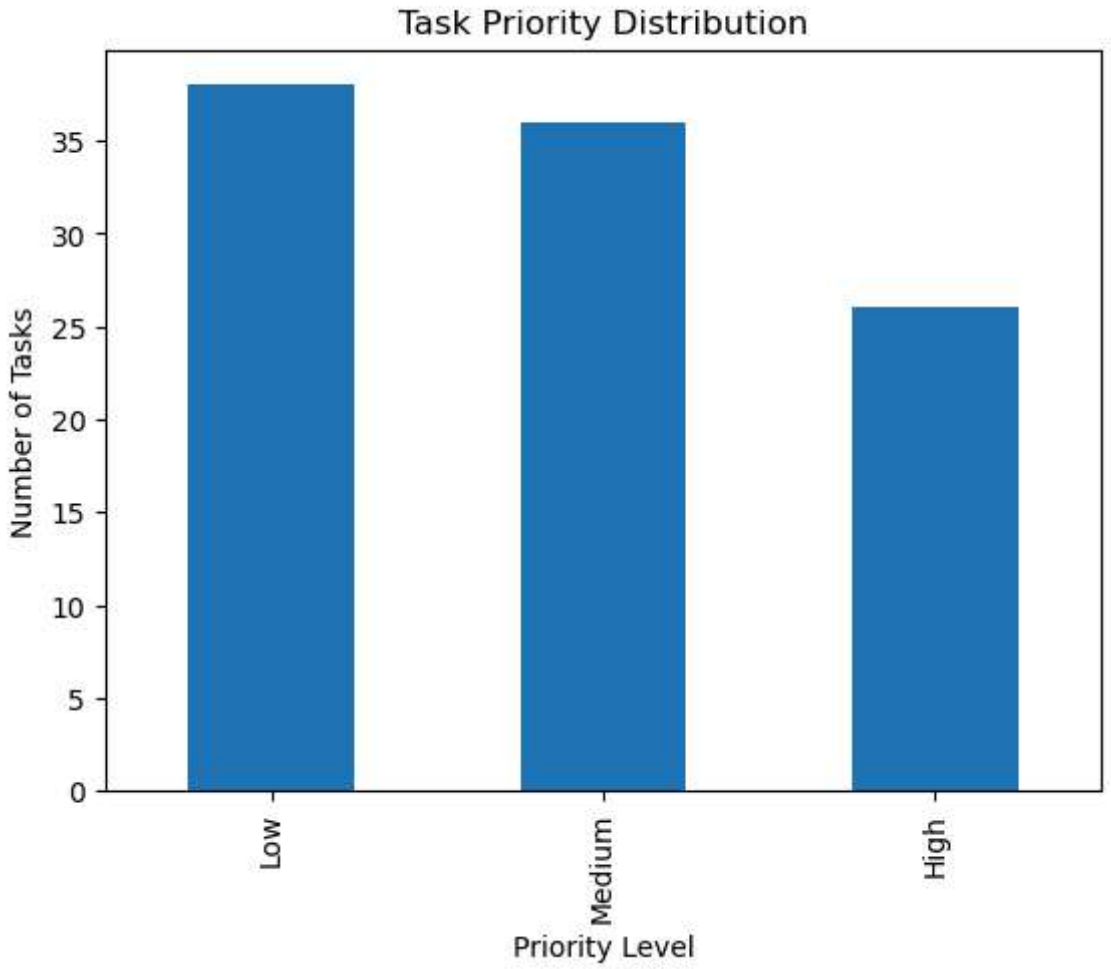


In [6]:
```
df["Priority"].value_counts()
```

Out[6]:
```
Priority
Low       38
Medium    36
High      26
Name: count, dtype: int64
```

In [7]:
```
import matplotlib.pyplot as plt

df["Priority"].value_counts().plot(kind="bar")
plt.title("Task Priority Distribution")
plt.xlabel("Priority Level")
plt.ylabel("Number of Tasks")
plt.show()
```



In [8]:
```
total_tasks = len(df)
completed_tasks = (df["Status"] == "Done").sum()
completion_rate = (completed_tasks / total_tasks) * 100

print(f"Total Tasks: {total_tasks}")
print(f"Completed Tasks: {completed_tasks}")
print(f"Task Completion Rate: {completion_rate:.2f}%")
```

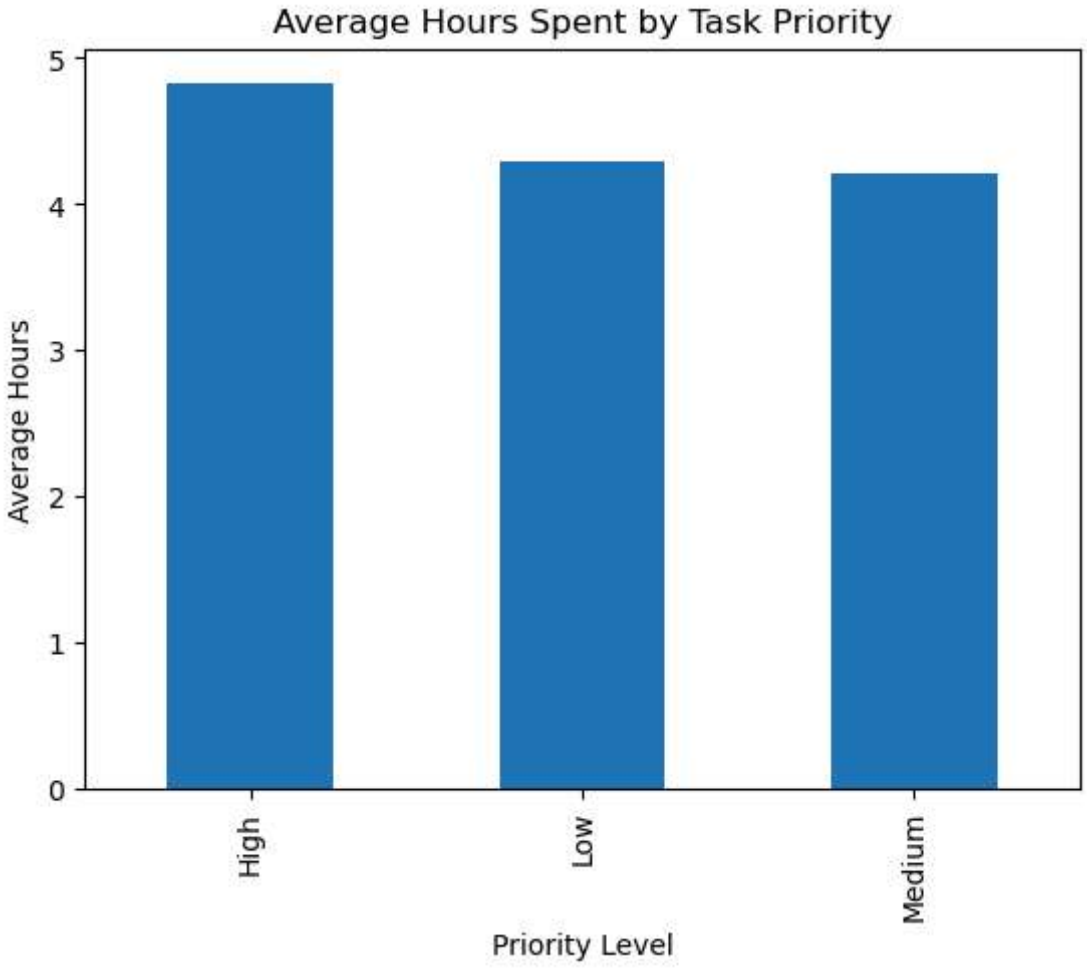
```
Total Tasks: 100
Completed Tasks: 22
Task Completion Rate: 22.00%
```

In [9]:
```
df.groupby("Priority")["Hours_Spent"].mean()
```

Out[9]:
```
Priority
High      4.823077
Low       4.294737
Medium    4.213889
Name: Hours_Spent, dtype: float64
```

In [10]:
```
df.groupby("Priority")["Hours_Spent"].mean().plot(kind="bar")
plt.title("Average Hours Spent by Task Priority")
```

```
plt.xlabel("Priority Level")
plt.ylabel("Average Hours")
plt.show()
```



Project Insights

This project analyzes a dataset of employee tasks.
The dataset contains information about task status, priority level, and hours spent.

Key Observations:

- A large number of tasks are marked as Delayed or Pending.
- High priority tasks require slightly more time on average.
- The task completion rate helps evaluate overall productivity.

This analysis helps understand workflow efficiency and task management patterns.

In [ ]: